Supporting Information

Accept or abstain? A pre-specified test of reliability gates calibrated on held-out chemistry for molecular property and reaction-yield models

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S1 Results of the MLP ensemble (M2)

Table S1. Ungated accuracy of M2 (mean ± s.d. over five seeds). Error rate: share of predictions with an absolute error above 0.5 log units (logP) or 10 percentage points (yield).

Test set	n	RMSE	R^2	Error rate
LogP, random R	2,449	0.62 ± 0.01	0.87 ± 0.01	0.30
LogP, held-out cluster	3,551	0.84 ± 0.02	0.74 ± 0.02	0.45
LogP, prospective E1	327	0.63 ± 0.03	0.61 ± 0.03	0.41
LogP, fluorinated E2 (all)	183	0.70 ± 0.02	-0.42 ± 0.07	0.48
sp^3 C–F compounds	140	0.70 ± 0.03	-0.70 ± 0.12	0.50
parents	43	0.70 ± 0.02	0.04 ± 0.06	0.42
BH Test1, random R	350	6.19 ± 0.45	0.95 ± 0.01	0.09
BH Test1, held-out additives	898	12.08 ± 0.95	0.80 ± 0.03	0.30
BH Test2, random R	351	6.15 ± 0.46	0.95 ± 0.00	0.10
BH Test2, held-out additives	900	12.69 ± 0.55	0.78 ± 0.02	0.34
BH Test3, random R	350	5.52 ± 0.36	0.96 ± 0.01	0.08
BH Test3, held-out additives	897	16.20 ± 0.99	0.67 ± 0.04	0.40
BH Test4, random R	350	5.47 ± 0.44	0.96 ± 0.01	0.08
BH Test4, held-out additives	900	17.87 ± 0.93	0.54 ± 0.05	0.48
SM, random R	576	10.91 ± 0.69	0.85 ± 0.02	0.25
SM, held-out ligands	960	19.48 ± 4.59	0.28 ± 0.61	0.45

Table S2. AURC of the selective error rate for M2 (lower is better; mean over seeds). "No gate" is the ungated error rate.

Held-out test	No gate	s_1	$-u$	Fused
LogP cluster	0.445	0.339	0.366	0.314
LogP prospective (E1)	0.408	0.292	0.373	0.284
LogP fluorinated (E2)	0.480	0.416	0.434	0.392
BH Test1	0.304	0.280	0.246	0.246
BH Test2	0.336	0.352	0.259	0.259
BH Test3	0.396	0.351	0.337	0.339
BH Test4	0.481	0.483	0.446	0.461
SM ligand	0.454	0.463	0.360	0.386

Table S3. Operating-limits card for Gate B with M2 at $t = 0.8$. Pooled precision with Wilson 95% intervals; other definitions as in Table 4 of the main text. The C_h-calibrated gate abstained on every query in every seed.

	Calibrated on C_r		Calibrated on C_h		
Held-out test	Coverage	Precision [95% CI]	Coverage	Precision [95% CI]	Seeds accepting
LogP cluster	0.07	0.82 [0.80, 0.84]	0.00	abstained	0/5
LogP prospective (E1)	0.08	0.81 [0.74, 0.87]	0.00	abstained	0/5
LogP fluorinated (E2)	0.06	0.69 [0.56, 0.80]	0.00	abstained	0/5
BH Test1	0.98	0.70 [0.69, 0.72]	0.00	abstained	0/5
BH Test2	0.97	0.67 [0.66, 0.69]	0.00	abstained	0/5
BH Test3	0.79	0.60 [0.59, 0.62]	0.00	abstained	0/5
BH Test4	0.99	0.52 [0.51, 0.54]	0.00	abstained	0/5
SM ligand	0.13	0.60 [0.56, 0.64]	0.00	abstained	0/5

S2 Operating-limits card of M1 at $t = 0.9$

Table S4. Operating-limits card for Gate B with M1 at $t = 0.9$. Definitions as in Table 4 of the main text.

	Calibrated on C_r		Calibrated on C_h		
Held-out test	Coverage	Precision [95% CI]	Coverage	Precision [95% CI]	Seeds accepting
LogP cluster	0.01	0.91 [0.84, 0.95]	0.00	0.92 [0.83, 0.97]	1/5
LogP prospective (E1)	0.01	0.92 [0.67, 0.99]	0.01	1.00 [0.72, 1.00]	1/5
LogP fluorinated (E2)	0.00	abstained	0.00	1.00 [0.34, 1.00]	1/5
BH Test1	0.45	0.91 [0.89, 0.92]	0.18	0.94 [0.93, 0.96]	3/5
BH Test2	0.61	0.87 [0.86, 0.88]	0.22	0.95 [0.94, 0.96]	4/5
BH Test3	0.34	0.85 [0.83, 0.87]	0.10	0.95 [0.92, 0.96]	3/5
BH Test4	0.70	0.65 [0.63, 0.66]	0.19	0.72 [0.69, 0.75]	2/5
SM ligand	0.27	0.88 [0.86, 0.90]	0.04	0.90 [0.85, 0.93]	1/5

S3 Analysis plan

The plan below was written before the evaluation and is reproduced in its original wording. Notes on its implementation follow at the end.

S3.1 1. Question

Can an accept-or-abstain gate, calibrated on data held out *by chemistry*, keep the precision of a molecular model's accepted predictions at a stated target when the queries come from chemistry the model has not seen? And does combining an input-distance signal with a model-confidence signal rank errors better than either signal alone?

S3.2 2. Definitions fixed in advance

- **Correct prediction.** LogP: $|y - \hat{y}| \leq 0.5$ log units. Reaction yield: $|y - \hat{y}| \leq 10$ percentage points.
- **Coverage** is the share of queries the gate accepts. **Selective precision** is the share of accepted queries that are correct. **Selective error rate** = 1 − selective precision.
- **Target precisions:** t = 0.8 and t = 0.9. Risk level for the threshold guarantee: $\delta = 0.1$.

S3.3 3. Data (all public)

- **D1 LogP pool:** SangsterLogP main sheet (Zenodo 10.5281/zenodo.19387551, CC BY 4.0). Parsed with RDKit; unparsable SMILES dropped; duplicates by InChIKey merged (first record kept). Every compound whose InChIKey occurs in either external set below is removed from the pool.
- **E1, prospective external set:** the SangsterLogP "Prospective" sheet (327 compounds).
- **E2, fluorinated external set:** the 183 measured amide logP values of `blackthorn-ai/fluoricLogPpKa` (MIT), each fluorinated compound beside its non-fluorinated parent.
- **D3a Buchwald–Hartwig** (Ahneman et al. 2018; as redistributed in `SoftServeInc/yield-paper`, MIT): 3,955 reactions; the four published out-of-sample additive tests (sheets Test1–Test4, training rows 1–3057, 1–3055, 1–3058 and 1–3055 respectively) and random splits.
- **D3b Suzuki–Miyaura** (Perera et al. 2018; same redistribution): 5,760 reactions, target `Product_Yield_PCT_Area_UV`.

S3.4 4. Representations and base models

- Molecules: ECFP4 counts (radius 2, 2,048 bins) plus ten RDKit descriptors (MolWt, TPSA, NumHDonors, NumHAcceptors, NumRotatableBonds, RingCount, NumAromaticRings, FractionCSP3, HeavyAtomCount, NumHeteroatoms). Crippen MolLogP is deliberately excluded.
- Reactions: concatenated ECFP4 bit vectors (1,024 bits) of each variable component (BH: aryl halide, ligand, base, additive; SM: reactant 1, reactant 2, ligand, reagent, solvent). Component structures follow the cited redistribution.
- **M1 (primary):** random forest, 300 trees, max_features = 0.2, other settings at scikit-learn defaults. Spread u = standard deviation of the per-tree predictions.
- **M2 (secondary, robustness only):** ensemble of five multilayer perceptrons (512–128, ReLU, Adam, L2 = 1e-4, early stopping on 10% of the fit set). Spread u = standard deviation across members.
- Hyper-parameters are fixed here and are not tuned on any calibration or test fold.

S3.5 5. Splits (five seeds, 0–4)

For each dataset a pool is divided by a group label: Butina clusters of ECFP4 at Tanimoto distance 0.65 (D1), additive (D3a), ligand (D3b).

- **Held-out test H:** D1: 15% of clusters; D3a: the published TestN additives; D3b: two ligands per seed.
- **Held-out calibration C_h:** groups disjoint from H and from the fit set. D1: 15% of clusters; D3a: four additives drawn from the TestN training additives; D3b: two further ligands.
- **In-distribution part I** (all remaining groups) is split at random into the fit set F (70%), the random calibration fold C_r (15%) and the random test R (15%).
- The base model is trained on F only. Both gates (C_r-calibrated and C_h-calibrated) use the same model.
- **D1 external tests:** the D1 models are also scored on E1. A separate D1 model, trained after removing every compound with an sp^3 C–F bond (SMARTS `[CX4]F`) from the pool, is scored on E2; fluorinated compounds and parents are reported apart.

Held-out tests are therefore: D1 cluster, D1–E1, D1–E2 (D1 family); D3a Test1–Test4 and D3b ligand (D3 family).

S3.6 6. Signals

- **s1:** maximum Tanimoto similarity of the query fingerprint to F; **s5:** mean of the five largest.
- **u:** ensemble spread of the base model.
- **Conformal intervals (nominal 90%):** normalised split conformal, score $|y - \hat{y}| / (u + \varepsilon)$, $\varepsilon = 0.1 \times \text{median}(u)$ on the calibration fold.

S3.7 7. Gates

- Each calibration fold C is split at random into halves C_a and C_b.
- **Single-signal gates:** score s1, or $-u$.
- **Gate B (fused):** logistic regression of correctness on (s1, s5, log u), fitted on C_a. Its score is the predicted probability of a correct prediction.
- **Threshold by Learn-then-Test** on C_b, for every score: candidate thresholds are the score quantiles of C_b, tested in fixed sequence from the strictest that accepts at least 50 points of C_b, loosening one step at a time. A threshold passes when the binomial p-value $P(\text{Bin}(n, t) \geq k) \leq \delta$, where n points are accepted and k are correct. The loosest threshold reached before the first failure is used. If none passes, the gate abstains on every query.
- **Gate A (conjunctive, no guarantee):** accept if $s1 \geq \tau_1$ and $u \leq \tau_2$; (τ_1, τ_2) from a 20 × 20 grid of quantiles of the whole C, chosen as the largest coverage whose empirical precision on C is at least t.

S3.8 8. Metrics

Per test set, model and seed: RMSE, MAE and R^2 ungated; error rate ungated; risk–coverage curves of selective error rate and selective RMSE for each score; AURC of the selective error rate (mean over all coverage levels k/n); coverage and achieved precision of every gate at t = 0.8 and 0.9; empirical coverage of the conformal intervals on all queries and on accepted queries. Results are averaged over seeds; 95% intervals come from 1,000 bootstrap resamples of test queries, pooled over seeds.

S3.9 9. Hypotheses and pass criteria (model M1)

- **H1, the gate helps where the model is weak.** On every held-out test, the selective error rate of the fused score at 30% coverage is lower than the ungated error rate, with the 95% bootstrap interval of the difference excluding zero.
- **H2, the calibration split matters.** Over the (held-out test × target) cells, Gate B calibrated on C_r falls short of the target by more than 0.05 in at least half of the cells; calibrated on C_h, it reaches at least t − 0.05 in at least three quarters of the cells (seed-averaged precision).
- **H3, the combination adds something.** The AURC of Gate B's score (C_h-fitted) is at least 10% lower, relatively, than the better single signal on at least two held-out tests in each of the D1 and D3 families.
- **H4, conformal coverage under shift.** Intervals calibrated on C_r cover at least 0.88 of R; cover less than 0.85 on at least one held-out test; and, on each held-out test where they fall below 0.85, cover at least 0.88 of the queries accepted by Gate B (C_h, t = 0.8).

Every hypothesis is reported whether or not it holds.

S3.10 10. Exclusions

No client, proprietary or confidential data, code or results are used. Biological context (gene expression, perturbation or cell-line shift) and generative models are out of scope.

S3.11 Notes on implementation

- A gate that abstains on every query has no defined precision. Such seeds are left out of the seed average. A cell in which every seed abstains counts as meeting the target in H2.
- H4 cannot be checked on a test where Gate B accepts nothing; this counts as a failure of H4 for that test.
- The held-out tests for H1–H4 are the logP cluster test, E1, E2 (all 183 compounds), BH Test1–Test4 and the SM ligand test.
- Holding out four calibration additives leaves about 1,640 reactions in F for each BH test. Ungated BH accuracy is therefore not directly comparable with published results on the same tests.
- The logP held-out sets were formed by assigning whole clusters until 15% of the compounds was reached.
- The Suzuki–Miyaura component structures come from a table written for this study (`src/aoa/data.py`). The table omits spectator fragments: the internal standard, HCl and water of hydration.
- For test sets that are identical in every seed (E1, E2, BH Test1–Test4), bootstrap intervals use the same resample for all seeds.

S4 Software environment

Python 3.12.14; RDKit 2026.09.1; scikit-learn 1.9.1; NumPy 2.5.4; SciPy 1.18.1; pandas 3.1.0; matplotlib 3.11.2. The full pipeline (`fetch_data.py`, `run_predictions.py`, `analyse.py`, `make_figures.py`, `make_tables.py`) runs in about 40 minutes on an 18-core laptop CPU. Input digests (SHA-256) are listed in `PROVENANCE.md` and checked by `fetch_data.py`.